

Lecture 4: Time Inconsistency & Present Bias

Why We Make Promises We Cannot Keep

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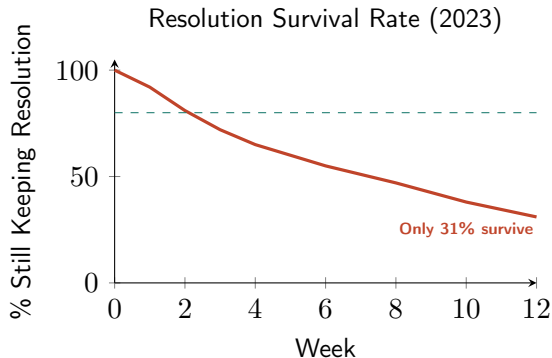
The New Year's Resolution Problem

The Statistics

- **80%** of New Year's resolutions fail by February (Statista, 2023).
- **50%** of gym memberships purchased in January are unused by March.
- **47%** of Canadians who intend to start saving "this year" have not started by December (FP Canada, 2023).

The Question

Is this a character defect? Lack of willpower?
Or is it a predictable feature of how we discount time? **Today we show it is the latter.**



Lecture Roadmap

1. **Exponential discounting** — the standard model and its properties
2. Evidence that exponential discounting fails — declining discount rates
3. Hyperbolic discounting — the alternative
4. The β - δ model — quasi-hyperbolic discounting
5. Procrastination — why we delay even when we shouldn't
6. Naivety vs sophistication — do we know we'll procrastinate?
7. Commitment devices — solutions for sophisticated agents
8. Applications: savings, addiction, smoking, gambling
9. Policy implications: sin taxes, defaults, mandatory savings

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The Model

A rational agent with discount factor $\delta \in (0, 1)$ values future utility as:

$$U_t = \sum_{\tau=0}^{\infty} \delta^{\tau} u(c_{t+\tau})$$

Each period discounted at a **constant** rate: $r = 1/\delta - 1$.

Three Reasons to Discount

1. **Opportunity cost:** \$100 today can earn interest.
2. **Uncertainty:** The future may not arrive.
3. **Pure time preference:** Impatience for its own sake.

Key Property: Time Consistency

If on **Monday** you prefer \$110 on Friday over \$100 on Thursday, then on **Thursday** you should *still* prefer \$110 tomorrow over \$100 today. The ranking of future options **does not change** as they approach. “Self₀ and Self_t always agree.”

Thaler's Evidence — Declining Discount Rates

Thaler (1981) — Experimental Method

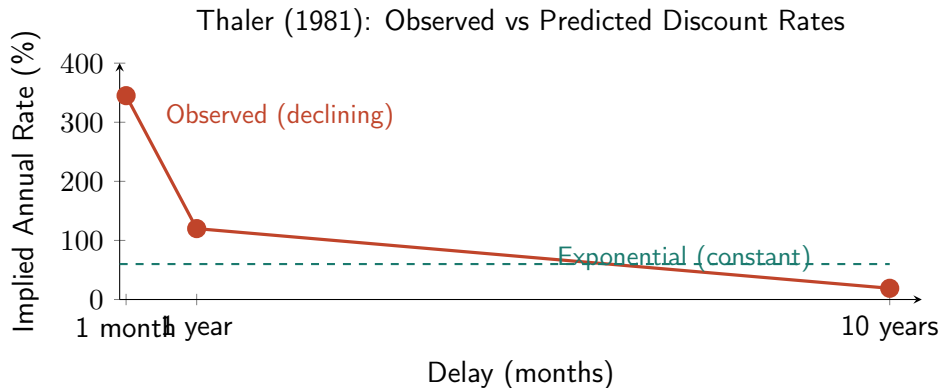
Participants asked: “What amount of money in the future would make you indifferent to receiving \$15 today?”

Amount now	Delay: 1 month	Delay: 1 year	Delay: 10 years
\$15 now → \$X later	Median: \$20	Median: \$50	Median: \$100
<i>Implied annual rate</i>	<i>345%</i>	<i>120%</i>	<i>19%</i>

The Violation of Exponential Discounting

Exponential discounting predicts a **constant** annual discount rate regardless of horizon. But the data show rates *falling sharply* from 345% (short horizon) to 19% (long horizon). This “**decreasing impatience**” cannot be explained by any constant-rate model.

Declining Discount Rates — The Evidence Graphically



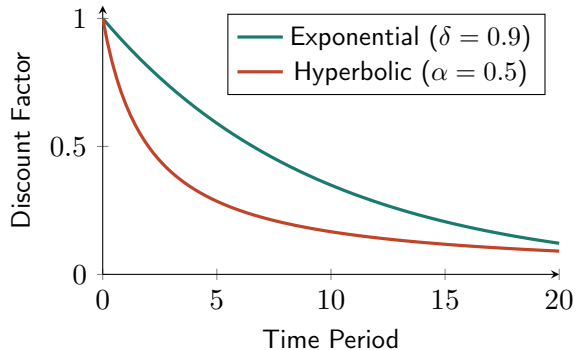
Key Insight

The steep drop in the short run (345% $\rightarrow$ 120%) followed by a flat tail is the signature of **hyperbolic discounting** — the model we study next.

Hyperbolic Discounting

Instead of δ^t (constant decay), use: $D(t) = \frac{1}{1 + \alpha t}$. This generates a **discount rate that falls over time** — matching the empirical evidence.

Exponential vs Hyperbolic



The Key Difference

Steep initial drop: present overweighted vs near future. **Flat tail:** future periods roughly equally valued. This initial drop *is* present bias.

Dynamic Inconsistency: Preferences **reverse** as time passes — you prefer to exercise tomorrow over today, but tomorrow you prefer the day after. Ad infinitum.

The β - δ Model (Quasi-Hyperbolic)

$$U_t = u(c_t) + \beta \sum_{\tau=1}^{\infty} \delta^{\tau} u(c_{t+\tau})$$

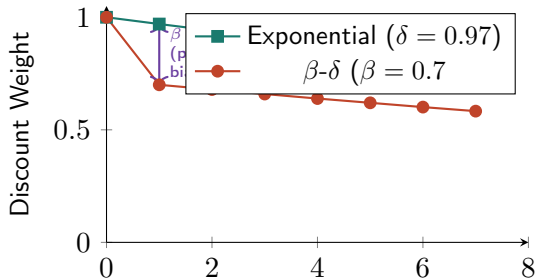
$\beta \in (0, 1)$: **present bias** (extra future discounting). δ : long-run patience. $\beta = 1$: reduces to exponential.

Calibrated Parameters: $\beta \approx 0.7$ – 0.8 ;

$\delta \approx 0.95$ – 0.99 .

$\beta = 0.7$: tomorrow is worth only **70%** as much as today, *regardless of how far away tomorrow is*.

β - δ vs Exponential ($\beta = 0.7, \delta = 0.97$)



Why We Procrastinate

O'Donoghue & Rabin (1999, 2001)

A present-biased agent faces a task: cost c today, benefit b in the future. The agent does the task *today* if:

$$u_{\text{today}}(\text{do}) \geq u_{\text{today}}(\text{wait})$$

With present bias: doing today has cost c (no discounting), benefit b discounted by β . Waiting has cost c discounted by β (since it's in the future). When b is far in the future, the agent keeps preferring to wait — every day.

The Procrastination Trap

Monday: “I’ll clean my room on Saturday” (Saturday cost is discounted by β).

Saturday morning: “I’ll clean tomorrow” (tomorrow’s cost is again discounted by β).

Same logic applies every period. The agent never cleans.

[GAME] The Homework Problem

Scenario

You have a take-home essay due at the **end of term** (6 weeks away). You can write it any time. It takes 3 hours.

The benefit (grade) arrives at end of term regardless of when you write it.

Poll: When do you plan to write it?

A) This week B) In 2–3 weeks C) The week before D) The night before

Second Poll

Honestly: How many of you did your most recent assignment *earlier than 48 hours before the deadline*?

The Gap

The gap between *intended* timing (this week) and *actual* timing (night before) is exactly what the β - δ model predicts. This gap is **present bias in action**.

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O'Donoghue & Rabin (2001)

Present-biased agents can differ in their *self-knowledge*:

Naïve Agent ($\hat{\beta} = 1$)

Does **not know** they are present-biased.

Believes their future self will follow through on plans. Perpetually surprised by their own failure to act.

"I'll definitely start my diet next Monday."

(Said every Monday for a year.)

Does not use commitment devices — sees no need.

Sophisticated Agent ($\hat{\beta} = \beta$)

Knows they are present-biased. Anticipates future self-control failures. May use commitment devices NOW to constrain the future self.

Ulysses and the Sirens: tied himself to the mast before hearing the sirens.

Ariely & Wertenbroch (2002): students chose self-imposed deadlines for papers — and performed better.

Evidence on Procrastination

Ariely & Wertenbroch (2002): Three MIT paper configurations:

1. **Imposed evenly spaced:** best grades.
2. **Self-imposed evenly spaced:** second best.
3. **All at end:** worst grades.

Lesson 1: Deadlines Work

Self-imposed deadlines improve performance.
Present-biased agents benefit from
commitment to milestones.

Lesson 2: Partial Sophistication

Students *chose* to space deadlines,
showing awareness of procrastination —
but many clustered too tightly at the end.

UBC (2019): Digital calendar reminders $\Rightarrow$ +0.4 GPA points vs controls. Technology as a commitment device.

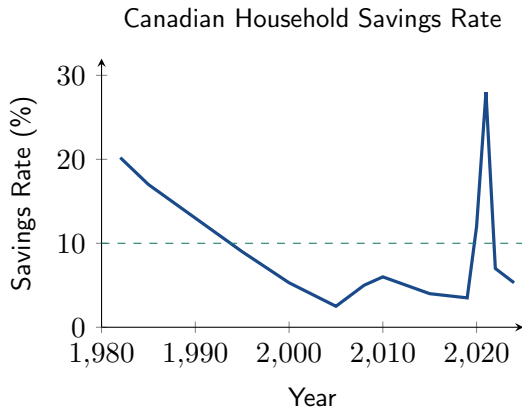
The Undersaving Puzzle — Canadian Data

Savings Rate (Statistics Canada):

- 1982: **20.2%** → 2005: **2.5%** (historic low)
- 2020–21: **12–28%** (COVID) → 2024: **≈5.3%**

The RRSP Gap

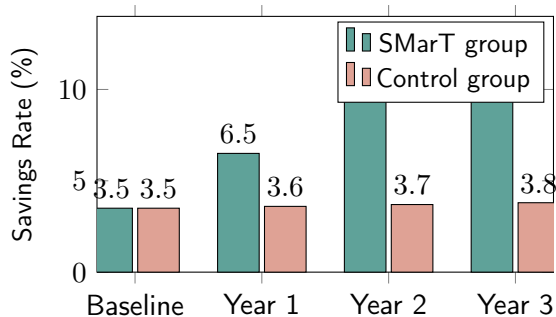
- Only **27%** contributed (2022); unused room: \$25K+.
- **41%** have *no* retirement savings (BMO, 2023).
- 93% *know* RRSPs exist. The problem is present bias.



SAVE MORE TOMORROW (SMarT)

Thaler & Benartzi (2004): Commit TODAY to save more starting with your **next pay raise**:
(1) commitment before temptation; (2) raise offsets loss aversion; (3) automatic.

SMarT Program Results



The Results

- Savings rate: 3.5% → 11.6% in 3 years.
- Participation: 3.5% → 13.6%.
- Control group: essentially unchanged.
- US Pension Protection Act (2006) institutionalised this nationwide.
- **British Columbia** WorkSafeBC pension: automatic escalation adopted 2018.

Definition

A **commitment device** is a mechanism that a sophisticated present-biased agent uses *now* to constrain their future self. It makes the future temptation either unavailable or costly.

Historical Examples

- **Ulysses and the Sirens:** tied to the mast before the temptation arrived.
- **Christmas clubs:** low-interest savings accounts locked until December.
- **Layaway plans:** commit to purchase via installments.
- **SEED accounts (Philippines):** Ashraf, Karlan & Yin (2006) — savings 82% higher than control after 12 months.

Modern Examples

- **stickK.com:** pledge money against a goal. If you fail, the money goes to a charity you hate. $\approx$ \$700K/day in active stakes.
- **Automatic savings transfers:** Simplii Financial (Canada), Chime.
- **Gym contracts:** pre-pay for a year to make yourself go.
- **Smoking bans at home:** remove availability

[GAME] Design a Commitment Device

Group Exercise — 4 Minutes

Your friend says: “I know I should exercise 3 times a week, but I keep skipping the gym when I’m tired. I’ve been saying this for 6 months.”

Design a commitment device that:

1. Works even when your friend is tired and unmotivated.
2. Does not require strong willpower in the moment.
3. Your friend cannot easily circumvent.

Evaluation Criteria

- Does it address *when* costs and benefits occur?
- Does it involve a *pre-commitment* before the temptation?
- Is it *credibly binding*? Can future-self escape?

Addictive goods: immediate payoff + delayed cost. Present bias ($\beta < 1$) causes systematic *over-consumption* because future costs are heavily discounted.

Canadian Smoking Data

- 13% smoke (2019) — down from 50% in 1966.
- 70% *want* to quit; 50% try/year; only **5%** succeed without aids.
- “I’ll quit Monday” — said every week, for years. Classic naive present bias.
- BC “QuitNow”: 23% 6-month quit rate vs 5% unassisted.

Policy: Corrective Taxes

O’Donoghue & Rabin (2006): if $\beta < 1$ and agents are *naive*, a Pigouvian tax is optimal even with *perfect information* — over-consumption relative to own long-run preferences.

Discussion Questions

- Q1. “Are you a sophisticated or naïve present-biased agent? How do you know? Can you think of a commitment device you’ve successfully used?”
- Q2. “Is it paternalistic for the government to require mandatory retirement savings (like Australia’s Superannuation at 11%)? Or is it correcting a cognitive bias that people themselves want to correct?”
- Q3. “Should cigarette taxes be *higher* if people are naïvely present-biased? How does your answer change if people are *sophisticated*?”

Key Takeaways — Lecture 4

- ✓ **Exponential discounting:** constant rate, time consistent — self₀ and self_t always agree.
- ✓ **Evidence:** discount rates decline with horizon — violates exponential model.
- ✓ **Hyperbolic discounting:** steep initial drop + flat tail = present bias.
- ✓ **β - δ model:** $\beta < 1$ = present bias; δ = long-run patience.
- ✓ **Time inconsistency:** choices reverse as future becomes present. Preferences flip.
- ✓ **Naïve vs sophisticated:** naïves cycle through failed resolutions; sophisticates pre-commit.
- ✓ **Commitment devices:** work by constraining future self before temptation arrives.
- ✓ **Applications:** procrastination, undersaving, addiction, SMarT program.

Next: Lecture 5: Intertemporal Choice & Savings — Mental accounting, why we hold debt and savings simultaneously, and the Canadian retirement savings crisis.

Further Reading

- Laibson, D. (1997). Golden eggs and hyperbolic discounting. *QJE*, 112(2), 443–477.
- O'Donoghue, T. & Rabin, M. (1999). Doing it now or later. *AER*, 89(1), 103–124.
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